Example 3.9 The sum of thrice the first number, second number and twice the third number is 5. If thrice the second number is subtracted from the sum of first number and thrice the third we get 2. If the third number is subtracted from the sum of twice the first, thrice the second, we get 1. Find the numbers.

Solution Let the three numbers be $x,\ y,\ z$

From the given data we get the following equations,

$3x + y + 2z = 5$ ...(1) $\quad x + 3z - 3y = 2$ ...(2) $\quad 2x + 3y - z = 1$ ...(3)

$(1) \times 1 \Rightarrow \quad 3x + y + 2z = 5$

$(2) \times 3 \Rightarrow \quad 3x - 9y + 9z = 6 \quad (-)$

$10y - 7z = -1$ ...(4)

$(1) \times 2 \Rightarrow \quad 6x + 2y + 4z = 10$

$(3) \times 3 \Rightarrow \quad 6x + 9y - 3z = 3 \quad (-)$

$-7y + 7z = 7$ ...(5)

Adding (4) and (5), $10y - 7z = -1$

$-7y + 7z = 7$

$3y = 6 \Rightarrow y = 2$

Substituting $y = 2$ in (5), $-14 + 7z = 7 \Rightarrow z = 3$

Substituting $y = 2$ and $z = 3$ in (1),

$3x + 2 + 6 = 5$ we get $x = -1$

Therefore, $x = -1,\ y = 2,\ z = 3$.

Thinking Corner

1. The number of possible solutions when solving system of linear equations in three variables are _____.
2. If three planes are parallel then the number of possible point(s) of intersection is/are _____.

Progress Check

1. For a system of linear equations in three variables the minimum number of equations required to get unique solution is _____.
2. A system with _____ will reduce to identity.
3. A system with _____ will provide absurd equation.

Exercise 3.1

1. Solve the following system of linear equations in three variables

(i) $x + y + z = 5;\ 2x - y + z = 9;\ x - 2y + 3z = 16$

(ii) $\frac{1}{x} - \frac{2}{y} + 4 = 0;\ \frac{1}{y} - \frac{1}{z} + 1 = 0;\ \frac{2}{z} + \frac{3}{x} = 14$

(iii) $x + 20 = \frac{3y}{2} + 10 = 2z + 5 = 110 - (y + z)$

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